From Coursework to Production Systems: Building Applied Data Science Capability Through Work, Reflection, and System Design

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Author's Note on Confidentiality and AI Assistance

This portfolio is based on my original professional experience, graduate coursework, and public-safe reconstruction of system design patterns from work I performed in an operational data environment. To protect proprietary and confidential information, system names, internal data, exact financial values, customer or vendor identifiers, operational screenshots, and protected process details have been removed, generalized, or replaced with descriptive labels.

AI tools were used as writing, structuring, editing, and reflection assistants during the development of this portfolio. AI was used to help organize ideas, refine language, compare framing options, and improve clarity. I reviewed, selected, revised, and approved the final content. AI output was not treated as an independent factual source for operational claims, and the systems described remain grounded in my own work, coursework, and professional judgment.

Introduction

This portfolio examines the relationship between my graduate coursework in Applied Data Science and the production-facing systems I built in my professional role as a Data Analyst. The central argument of this paper is that my coursework and professional work did not develop separately. They formed a feedback loop. My work gave the coursework operational context, and the coursework gave my work stronger language, structure, and methodology.

Often, when describing this relationship, I simply say:

"My work informs my degree, and my degree informs my work."

My path to applied data science did not begin in a classroom. It began in service. I spent 14 years in the United States Air Force working in mission-critical environments where systems had to function under pressure and where unclear information could have fatal consequences. That experience shaped how I think. It taught me to value precision, brevity, traceability, and operational usefulness. It also taught me that information is only valuable when it reaches the right person, in the right form, at the right time.

Transitioning from military life into civilian life required me to rediscover who I was outside the uniform. That transition was difficult, yet deeply meaningful. In the process, I discovered my passion for data as my bridge to understanding systems, reducing ambiguity, and solving real human-centered problems. Service did not end when I left the military. It changed form. Today, I serve through systems: by building tools that make work easier, decisions faster, and information more trustworthy. This purpose has become my organizing principle.

While completing my M.S. in Applied Data Science, I was also working inside a live operational environment where data problems were not theoretical. Reports affected planning decisions. Reconciliation logic affected financial review. Compliance workflows affected accountability. Inventory visibility affected operational risk. In that environment, the concepts I learned in class were immediately tested against real constraints: incomplete data, fragmented source-systems, stakeholder expectations, manual workarounds, process ownership, governance, and the need to make information useful to people who did not have time, nor the domain knowledge to reconstruct the logic behind every number.

This paper analyzes how I changed as a professional, going beyond a simple summary of my academic coursework. Across the program, I moved from thinking primarily in terms of reports, dashboards, and individual analyses toward thinking in terms of systems. A report answers a question. A system changes how questions are asked, answered, repeated, reviewed, and trusted. That distinction is also reflected in decision-support and data governance literature, which emphasizes that data systems create value when they improve trust, decision quality, and use rather than simply producing isolated outputs.[1] A mature system goes one step further by: 1)

removing recurring questions from the workflow by making answers visible, 2) self-service, and/or 3) structurally available at the moment they are needed. Much of my work has been focused on this shift. Instead of answering stakeholder questions one by one, I build systems that remove entire categories of questions from the workflow.

The portfolio is organized around three sanitized, public-facing case studies drawn from operationally deployed systems I built through my professional work:

1. Financial Control System
2. Chemical Compliance System
3. Inventory Control System

Because these case studies are drawn from real production work, this paper uses generalized names and sanitized descriptions to protect proprietary information. Their purpose is to show how graduate-level concepts became professional capability. Each case study represents a different operational domain, but the underlying pattern is consistent: 1) define the problem, 2) structure the data, 3) clarify the workflow, 4) preserve evidence, and 5) design for human use.

The Financial Control System represents my longest and most complex development arc. It began as a reconciliation problem and developed into a structured control workflow for price-change review, variance analysis, and stakeholder communication. The Chemical Compliance System represents a compliance and governance workflow, where the main challenge was preserving reviewability, sign-off logic, and audit defense. The Inventory Control System represents an operational intelligence workflow, where the main challenge was transforming time-sensitive inventory data into a prioritized control surface that helped users understand what needed attention first.

These projects matter because they show applied data science operating in its real environment. In coursework, data is often bounded by an assignment prompt. In operational reality, data is surrounded by people, systems, incentives, legacy processes, missing context, and organizational risk. My graduate program helped me develop the tools to work inside that complexity without reducing it to a simple technical task.

Literature Review: Decision Support, Governance, and Production-Grade Tradeoffs

Although the systems discussed in this portfolio emerged from professional practice rather than from a formal research design, they can be interpreted through several established bodies of literature: 1) data governance, 2) decision-support systems, 3) workflow integration, 4) continuous monitoring, and 5) perishable inventory management. These areas are useful because they explain why applied data science work must extend beyond producing reports or isolated

analytical outputs. In deployed environments, data systems must support trust, reviewability, reproducibility, accountability, and action. The literature therefore provides a comparison frame for understanding the three case studies in this portfolio. It does not claim that the systems were built directly from these sources; rather, it shows that the patterns developed through practice align with broader research on trustworthy and operationally useful data systems.

Research on data governance is especially relevant to the central argument of this portfolio. Brous and Janssen argue that data science decision outcomes are not always adopted by decision-makers, often because users are uncertain about the quality of the data input. They position data governance as a way to improve trust in data science decision outcomes.[7] This connects directly to the systems in this portfolio because each project involved more than producing a final number or report. The Financial Control System, Chemical Compliance System, and Inventory Control System all required users to trust where the information came from, how it was structured, and whether it could be reviewed later. Abraham, Schneider, and vom Brocke similarly define data governance around authority and control over data management, with the purpose of increasing data value while reducing data-related cost and risk.[2] Their framework helps explain why the systems in this portfolio functioned as lightweight governance layers rather than simple dashboards.

Decision-support literature also supports the distinction this paper makes between reports and systems. A report answers a question once, while a decision-support system helps users act repeatedly under changing conditions. This is important because the portfolio systems were not designed merely to display information. They were designed to structure review, prioritize exceptions, preserve evidence, and guide human action. In that sense, they functioned as operational decision-support surfaces, for example, 1) the Financial Control System supported variance review and escalation, 2) the Chemical Compliance System supported review and sign-off, and 3) the Inventory Control System supported prioritization of expiration-sensitive inventory. The value came from embedding analysis into workflow rather than leaving it as a detached output.

The literature also complicates an overly optimistic view of decision-support systems. Decision-support tools can fail when users over-trust flawed outputs, when the system is not usable, or when recommendations are poorly integrated into the work people actually perform. Brauner and colleagues found that defective decision-support systems can reduce trust, perceived usefulness, acceptance, and overall performance.[3] Clinical decision-support research reaches a similar caution from a workflow perspective; inappropriate alerts can interrupt users, contribute to alert fatigue, and lead people to ignore or disable the support system.[4] These cautions are important because they explain why my systems were not designed as autonomous decision-makers. Their purpose was to organize evidence, surface exceptions, and make review easier while preserving human judgment at the point of decision.

Continuous monitoring and control literature provides a useful comparison for the Financial Control System. Alles, Brennan, Kogan, and Vasarhelyi describe a continuous monitoring layer built over an enterprise system with read-only interaction, showing how controls can be strengthened by monitoring business processes rather than only reviewing outcomes after the fact.[5] This comparison is useful because the Financial Control System operated as a local control layer around financial and operational evidence. However, the comparison also highlights a boundary. The goal of my system was not to fully automate audit judgment. Instead, it structured the evidence, classified exceptions, preserved traceability, and made human review more timely and consistent. This design choice reflects a practical tradeoff, automate structure where possible, but preserve human interpretation where judgment remains necessary.

The Inventory Control System also connects to research on perishable inventory management. Perishable inventory creates a time-sensitive decision problem because the value and risk of a record change as expiration approaches. Staff and Mustafee emphasize that perishable products have limited shelf life and that managing them requires decision support around uncertainty, issuing policies, and age tracking.[6] While my system was not a formal simulation model, it addressed a related operational need: helping users understand which inventory required attention first. The system used expiration-aware priority, exception detection, filters, and snapshot logic to move from raw visibility toward operational intelligence. The comparison clarifies the scope of my work. It was not an optimization engine, but it functioned as a practical review surface for time-sensitive inventory decisions.

Across these bodies of literature, a consistent theme emerges; production-grade data systems succeed when they connect information to trustworthy action. At the same time, the literature warns against over-automation, poor workflow fit, alert fatigue, and documentation that becomes burdensome rather than useful. These tradeoffs shaped the interpretation of my portfolio systems. The systems were intentionally lightweight, local, and human-reviewed. They were not enterprise platforms, autonomous decision engines, or formal research prototypes. Their value came from operating in the middle layer between fragmented source-systems and human decisions. They made source data, review status, exceptions, and evidence more visible while preserving the user's responsibility to interpret and act.

This literature review helps position the three case studies academically without replacing the experiential basis of the paper. The Financial Control System aligns with data governance, reconciliation, continuous monitoring, and decision-support literature. The Chemical Compliance System aligns with governance, review-state control, workflow accountability, and audit defensibility. The Inventory Control System aligns with decision support, perishable inventory management, prioritization, and workflow usability. The difference between the systems in this portfolio and the systems described in the literature is mainly one of scale and formality, not underlying concern. The literature often describes enterprise systems, clinical

systems, audit systems, or formal optimization models. My work describes public-safe, local, production-facing systems built under practical constraints. However, all of them address the same core question: how can data be structured to enhance a person's ability to consistently make better, faster, and more trusted decisions?

Course-to-Production Feedback Loop

The central pattern of this portfolio is a course-to-production feedback loop. My professional work exposed me to ambiguous operational problems, and my coursework gave me frameworks for understanding those problems more clearly. At the same time, the workplace gave the coursework weight. Concepts such as requirements, governance, schema design, change management, cloud architecture, variance analysis, and decision support became more than academic vocabulary. They became ways of naming problems I was already trying to solve.

This feedback loop was most visible in how my thinking changed over time. Earlier in my work, I often approached problems through the visible output: a report, a dashboard, an extract, or a calculation. Over time, my coursework pushed me to ask deeper questions. Who is the user? What decision does this support? What is the source of truth? What assumptions are hidden in the data? What happens if someone else has to maintain this process? Can the result be reviewed later? Does the workflow produce trust, or does it only produce another file?

These questions challenged how I built and, also, changed how I communicate. I began to see that providing numbers is only the baseline; a true data professional builds the conditions under which numbers can be trusted. Data governance research frames this kind of trust as a product of authority, control, quality, accountability, and risk management rather than data output alone.[8] That requires structure, documentation, repeatability, and an understanding of the human process around the data.

IST 654, Information Systems Analysis, was one of the strongest influences on this shift. The course gave me formal language for concepts I had already encountered at work but had not always named clearly. Requirements, use cases, ERDs, DFDs, scope boundaries, and functional decomposition helped me understand that many workplace problems were fundamentally system-definition challenges. If the users, boundaries, inputs, outputs, and assumptions were unclear, the final report could still be technically correct while failing the real need. That lesson appears throughout the Financial Control System, where the work required clarifying which records belonged in scope, how different source-systems related to one another, and how exceptions should be classified.

IST 621, Information Management and Technology, helped me understand the human side of the same problem. Digital transformation is often described through tools, platforms, and automation, but my experience showed me that transformation is also cultural. A new system

changes how people work, how they trust information, and how they understand their role in a process. In my own workplace, transformation did not begin with a large enterprise platform. It began with practical systems that made existing work clearer, faster, and easier to review. This course helped me see myself not only as someone who builds outputs, but as a change agent working inside a larger organizational transition.

IST 659, Data Administration and Database Management, strengthened the structural side of my work. It reinforced that reliable analysis depends on reliable data structures. Keys, constraints, schema design, insertion order, and repeatable queries may sound technical, but they also represent a professional discipline: the discipline of making sure that data relationships are explicit rather than assumed. This became especially important in the Financial Control System, where the idea of a truth table became central. The goal was to create a structured view where each record could be explained, reviewed, and traced back to its supporting logic.

IST 614, Information Technology Management and Policy, helped me place local systems inside enterprise strategy. Many of my workplace tools existed between formal enterprise systems and urgent local operational needs. That middle layer can be easy to dismiss as temporary or informal, but it is often where real work happens. IST 614 helped me understand that local innovation still needs governance, alignment, documentation, and strategic awareness. A tool that solves a local problem can either remain a workaround or become a bridge toward better enterprise architecture. That distinction shaped how I thought about my projects, especially when they touched compliance, financial controls, and future platform migrations.

IST 615, Cloud Management, helped me frame lightweight tools as platform decisions rather than isolated scripts. Many of my systems used accessible tools and low-friction delivery methods, but the architecture questions were still cloud questions: access, reliability, ownership, integration, cost, adoption, and governance. The lesson was that even small tools sit inside a platform context and should be designed with that context in mind.

SCM 651, Business Analytics, connected analysis to operational decision-making. It reinforced that analytics should support decisions under constraints. In the Inventory Control System, this was especially important. While inventory data existed, users needed help deciding what required attention first. That required prioritization, segmentation, and a review surface that translated raw data into action.

IST 686, Quantitative Reasoning for Data Science, strengthened my ability to evaluate claims and interpret uncertainty. It also shaped how I used AI in my work. I do not treat AI output as inherently correct. I use it to draft, reflect, compare, structure, and test ideas, while keeping deterministic logic in code, tables, and defined workflows. That distinction matters because several of my systems benefited from AI-assisted design and communication, but the actual reporting and review logic remained structured and reproducible.

Across these courses, the pattern became clear. My coursework gave me a way to think about work as systems. My workplace, in turn, gave me an environment where those systems had consequences. The result was a professional method that appears across all three case studies:

1. Context
2. Structure
3. Modularity
4. Evidence
5. Iteration

First, I identify the real operational context, then I structure the data and process boundaries. After which, I separate the work into modules so that logic, review, and output are not collapsed into one fragile process. Most importantly, I preserve evidence so that claims can be validated, and finally, I iterate based on feedback, use, and changing operational needs.

This is where the portfolio becomes more than a record of completed assignments. The systems I built are important, but the deeper evidence is the change in how I think. I became more careful about assumptions, more aware of governance, more attentive to users, and more focused on building workflows that other people could understand and trust. That is the difference between completing assignments and becoming an applied data science professional.

Financial Control System

This case study demonstrates my growth from analyst to systems designer by showing how reconciliation logic became a repeatable control workflow.

The Financial Control System is the strongest example of my course-to-production feedback loop because it began as a real operational problem and developed into a structured control workflow. This case study examines a financial and operational reconciliation system built around price-change review. The important point is the transformation the system represents: a fragmented reconciliation problem became a repeatable system for validating timing, classifying variance, preserving evidence, and making financial risk easier to explain.

Beneath incorrect prices or incomplete reports lay a deeper issue: different source systems described the same business process from different angles. Pricing records, invoice activity, operational movement, comparison baselines, and review logic did not automatically align. When those sources were reviewed separately, stakeholders could see pieces of the truth, but they could not easily see how those pieces fit together and that created a recurring cycle of reconstruction. Each time a question came up, the logic had to be rebuilt, re-explained, or re-validated.

This is where my coursework changed the way I approached the work. Before this project matured, I might have treated the problem primarily as a reporting challenge: pull the right data, calculate the variance, and create a useful output. That would have helped, but it would not have solved the structural issue. The more important question became: what system would make this review repeatable, explainable, and trusted? Continuous monitoring and business-process control literature supports this shift from after-the-fact review toward structured visibility over control processes, while also recognizing that audit judgment cannot always be fully automated.[9]

Systems analysis (IST 654) provided the discipline for that shift. In this project, defining the scope-boundaries mattered just as much as the calculation. A record could appear in one source and not another. A timing assumption could be valid for one population and invalid for another. A variance could represent a true process issue, a methodology difference, a chronology issue, or a source-data limitation. Without formal process decomposition, those categories could collapse into one ambiguous bucket called "mismatch." Defining the inputs, assumptions, and exception pathways before building the tables helped me resist that collapse.

The structural side of the project relied heavily on database management principles (IST 659). The system depended on the idea of a reconciliation table, or truth table, that could align operational and financial evidence in one reviewable structure. That table was not just a spreadsheet output. It was a schema decision. It required stable identifiers, defined relationships, transformation logic, and evidence fields that preserved why a record was classified the way it was. Concepts such as keys, constraints, source consolidation, and repeatable query logic helped me think about the table as a durable data object rather than a temporary analysis file.

The most important technical insight was the separation of concerns. Financial billing behavior and physical inventory behavior had to be modeled as distinct dimensions, but they also had to remain available inside the same review system. An invoice explains where the financial outcome appeared. A material movement explains what happened operationally. While those two facts may be related, treating them identically creates ambiguity. Separating them without recombining them, however, would also have been incomplete. The strength of the system came from a hybrid approach: it combined the discipline of a financial true-up audit with the operational insight of inventory consumption. One view could explain the financial result, while the other could explain the operational reality behind it. Together, they made the reconciliation model more powerful, more defensible, and more useful than either perspective would have been alone.

Business analytics (SCM 651) helped connect the mathematical variance to actual decision support. Beyond calculating the difference, the system helped stakeholders understand which exceptions required attention, which differences were methodological, which records were out of scope, and which issues had enough support to move forward. This changed the role of

analytics from passive explanation to active prioritization. The system had to support decisions under real constraints, including time, incomplete information, and different stakeholder needs.

From a governance perspective (IST 614), this project existed inside a larger enterprise context where local operational needs, enterprise systems, future platform migrations, and process ownership all intersected. The Financial Control System helped me see how local innovation can become a bridge instead of a workaround. By documenting logic, defining the review structure, and preserving traceability, the system created a stronger foundation for future enterprise requirements.

Quantitative reasoning (IST 686) strengthened the way I evaluated those claims. In this project, the numbers mattered, but the interpretation mattered just as much. A variance value without context can mislead. A high-risk category without source logic can create confusion. Quantitative reasoning helped me stay focused on evidence, assumptions, limitations, and defensible interpretation. That mindset also shaped how I used AI during the project. AI could help me draft language, test framing, organize logic, and think through presentation strategy, but the actual control logic had to remain structured, traceable, and reproducible. This caution aligns with decision-support research showing that erroneous systems can damage trust, perceived usefulness, acceptance, and performance when users rely on them without sufficient validation.[10]

Stakeholder design was just as important as the technical design. The system had to serve multiple audiences. Leadership needed the issue summarized clearly, finance stakeholders needed confidence in the reconciliation logic, and operations teams needed to understand the process implications. Analysts and process owners needed enough detail to review exceptions. The work had to extend beyond my own use to become legible, reviewable, and useful to others.

I describe the project as a Financial Control System because it created a repeatable way to ask, answer, review, and trust a family of related questions. It reduced the need for manual reconstruction by making the answer structure visible. It also created a foundation for self-service review, where stakeholders could understand not only the result but the logic behind it.

The strongest lesson from this project is that applied data science extends into control, trust, and translation. Analysis explains what happened, while a control system changes what happens next. Because this work developed through my Green Belt project, the DMAIC (Define, Measure, Analyze, Improve, and Control) framework also shaped the way I approached it. Define clarified the real problem and the stakeholders affected by it. Measure established the baseline and made the variance visible. Analyze helped separate source-system assumptions, timing issues, and true process gaps. Improve moved the work from explanation into a repeatable reconciliation workflow. Control became the most important phase because the goal was to prevent the same questions from needing to be reconstructed again, reduce the risk of continued financial leakage, and make future variance visible earlier. In this case, the work moved from

identifying variance to building a workflow that made variance review more structured, explainable, and actionable. That is exactly where my coursework and production work reinforced each other. The workplace gave me the problem and the coursework gave me the language and methods to structure it. The final system became evidence that the feedback loop was real.

Chemical Compliance System

This case study demonstrates my development in data architecture and governance by showing how a fragmented manual reporting process was engineered into a secure, audit-ready workflow.

The Chemical Compliance System shows a different side of my course-to-production feedback loop. Where the Financial Control System focused on reconciliation, variance, and control, this project focused on compliance, reviewability, and audit defensibility. The work began with a recurring reporting process, where the true challenge was making the process repeatable, reviewable, and able to withstand turnover, timing pressure, and later reconstruction.

Compliance work can look deceptively simple from the outside. A report is due, the data is gathered, the review is completed, and the requirement is satisfied. In practice, the risk sits in everything around the report: 1) who reviewed the records, 2) when they reviewed them, 3) what changed, 4) what was still pending, 5) which items required correction, and 6) whether the process could be reconstructed later. If that information is distributed across spreadsheets, emails, shared folders, and individual memory, the process becomes fragile even when the final report is submitted on time.

The process also had a recurring timing problem. The reporting requirement was known in advance, but the work historically compressed into a late-stage push near the deadline. That created pressure on the people gathering the data and raised a more important question: could the organization trust the numbers being assembled? Before the system matured, the process relied on snapshots, piecemeal records, and manual verification. The underlying issue was data maturity. We first had to establish what “right” looked like before the process could become more structured, governed, and repeatable.

That foundation took time. In the prior cycle, the work required heavy manual effort, including physical verification of inventory. I had to build confidence in the numbers by pulling and comparing reports regularly enough to understand the ground truth of the inventory picture. That groundwork changed what became possible the following year. Once the baseline was cleaner, I could move from manually reconstructing the reporting population to building a system around it. The project shifted from snapshots and emergency review toward a more active review process supported by live inventory visibility, structured sign-off, and traceable evidence.

The operational impact was visible in the timeline. In the earlier cycle, the work of gathering, reconciling, transforming, and preparing the information stretched across several months and still required an extended review period near the deadline. After the foundation was established, the next cycle compressed significantly. The preparation window moved from a long, fragmented effort into a focused December-to-January build, and the actual user review and entry process was designed around a one-week execution target. That mattered because the system reduced the amount of time spent searching for information, questioning which numbers were current, and manually piecing together records.

The user interface was part of that improvement. By replacing large, shared spreadsheets with a controlled review surface, the system guided users straight to the information they needed. This design choice also responds to workflow-centered decision-support research, which warns that poorly integrated support mechanisms can interrupt work, increase fatigue, and lead users to ignore or disable them.[12] In practical terms, the user workflow became much simpler: 1) review the line, 2) verify the information, 3) copy the prepared values when needed, 4) complete the external entry step, and 5) sign-off.

That design reduced unnecessary data handling and made the process easier to complete consistently. It also changed the nature of the work. Instead of spending most of the time searching, assembling, and reconciling information, reviewers could focus on validating and completing the required compliance actions. The project compressed both the readiness timeline and the execution timeline because the information was already structured, accessible, and prepared for use.

The main shift in my thinking was that compliance reporting is fundamentally a data architecture problem. The system needed to answer five questions clearly:

1. Who reviewed what?
2. When did they review it?
3. What changed?
4. What still required attention?
5. Could the process be reconstructed later?

Those questions shaped the system more than any single output did. The system had to create a workflow where review status, sign-off activity, threshold checks, and audit history could be understood as part of the same process.

Applying the requirements-gathering principles from IST 654 helped me frame this as a workflow problem rather than just a data extraction task. The users were not all asking for the same thing. Compliance coordinators needed process visibility; area managers needed a way to review and sign off, operational reviewers needed grouped records that made sense in context,

and process owners needed confidence that the review could be repeated. Mapping these distinct use cases allowed me to define the boundaries of the system, 1) the reporting population, 2) the review states, 3) the sign-off logic, and 4) the exception path.

The challenge was also cultural, which connected directly to information management and technology (IST 621). The system changed how people interacted with compliance information. It reduced reliance on individual memory and made the review process more visible to the people responsible for it. Digital transformation requires a movement from scattered work to shared understanding. In this case, the system helped turn a recurring compliance burden into a more coordinated and transparent workflow.

Cloud management concepts (IST 615) shaped how I thought about the delivery method. This system did not need to become a large enterprise deployment to provide value. It needed to be accessible, structured, governed, and usable by the people involved in the review process. Lightweight platform delivery mattered because the process lived in the space between local operational needs and larger enterprise systems. That middle layer is where many real compliance and operations problems exist. The system had to be practical enough to use immediately while structured enough to support future improvement.

Information technology policy (IST 614) helped me understand the data governance implications. Compliance systems demand accountability alongside efficiency. Data governance literature helps explain this requirement because compliance workflows depend not only on storing information, but also on authority, ownership, control, reviewability, and risk reduction.[11] Governance, in this case, meant designing the workflow so that responsibility and review history were not hidden inside informal communication.

Database administration (IST 659) supported the traceability side of the project. The system needed structured records, stable review states, sign-off events, and audit logging. Those are database concepts, but they also represent a way of thinking about accountability. A review action should leave a trace that can be understood later. That trace is what turns a one-time reporting effort into a defensible process.

The core design pattern was review-state control. Records needed to move through visible states such as pending review, reviewed, flagged, needs correction, complete, and archived. That state model made the process easier to manage because it showed what still required attention. By building review status directly into the workflow, the system reduced ambiguity and reliance on memory.

Another important design choice was grouped review. Compliance data can become difficult to review when related records are scattered across disconnected rows or files. Grouping related components into a reviewable unit reduced review fatigue and made the process easier to follow. This was both a usability improvement and a control improvement. When reviewers can

see related information together, they are more likely to make consistent decisions and less likely to miss context.

The system also needed to reduce the risk of overwrite and version confusion. Shared spreadsheet-based workflows are useful, but they can create uncertainty when multiple people interact with the same information. By controlling how review events were captured and logged, the system made the process more stable. The system was designed to support human review with enough structure that the workflow could be trusted.

Stakeholder design mattered here as much as it did in the Financial Control System. Compliance coordinators needed a summary of open items. Area managers needed a clear way to complete review responsibilities, operational reviewers needed enough detail to understand what they were signing off, and process owners needed audit defensibility. The system had to make the work legible to the people responsible for completing it, extending far beyond simple data storage.

This project also carried professional and reputational impacts. The Chemical Compliance System improved operational efficiency by making a recurring compliance-support process easier to review, track, and repeat. It reduced dependence on manual reconstruction and strengthened continuity. It also received internal operational efficiency recognition, which made it one of the clearest examples in my portfolio of coursework becoming production capability.

The strongest lesson from this project is that compliance work must outlast the person doing it. A process dependent on one person's memory is inherently unstable. A stronger system preserves lineage, ownership, reviewability, and accountability over time. That lesson connects directly to my broader development as an applied data science professional. I learned that a system's true value lies in whether the process can be trusted tomorrow.

Inventory Control System

This case study demonstrates my ability to design decision-support tools by showing how raw, expiration-sensitive data was transformed into an operational intelligence surface.

The Inventory Control System shows how my coursework helped me move from visibility to prioritization. In many operational environments, inventory data exists in some form. The harder problem is whether the people responsible for acting on that data can quickly understand what needs attention first. This project focused on expiration-sensitive inventory, where time itself changes the risk profile. A record that looks acceptable today may become urgent later. A static extract shows what exists without automatically telling users what to do next.

Even with available data, inventory review often becomes reactive when expiration-sensitive materials are spread across fragmented reports, source-systems, locations, and review processes. Users may know inventory exists but lack clarity on which inventory should be used first, which records should be monitored, which items require escalation, or which exceptions are emerging over time. Additionally, the sheer volume of data in a manual process causes users to miss information, make mistakes, lose context, miss timelines, or cut corners. That gap between visibility and action is where the system became valuable.

The key insight was simple:

1. Visibility shows the inventory.
2. Prioritization shows what needs attention.
3. Operational intelligence explains why.

That distinction shaped the system. The system organized the work around expiration-aware priority, exception detection, filtering, and review status to eliminate manual sorting. It moved the question from "What inventory do we have?" to "What inventory needs attention first, and why?"

Business analytics (SCM 651) was one of the strongest course connections for this project. The course reinforced that analytics should support decisions under constraints. In the Inventory Control System, the constraint was time. Expiration-aware inventory is fundamentally a decision-support problem. Research on perishable inventory management similarly emphasizes that limited shelf life, age tracking, uncertainty, and issuing policies make inventory review a time-sensitive decision problem rather than a static reporting task.[13] The system needed to help users act before risk became waste, delay, or operational disruption. That required prioritization logic, segmentation, and a way to communicate risk clearly.

IST 654 was critical here for defining the user workflow. Different users needed different views of the same truth. Operational leaders needed a summary of risk and open issues. Material handlers needed record-level clarity. Planners needed filtering and segmentation. Process owners needed a way to understand trends and follow-up actions. Structuring the system around those specific requirements prevented the application from becoming a generic, flat dashboard.

Understanding the adoption side of the project was guided by IST 621. A system succeeds when people use it to change how work gets done. The Inventory Control System was valuable because it made the review process easier to understand. It created a shared surface where users could see priority bands, exception categories, and relevant filters without needing to reconstruct the logic themselves. That is digital transformation at a practical level: helping people coordinate around better information.

Quantitative reasoning (IST 686) influenced the way I thought about time-based risk and interpretation. Expiration-aware review requires careful reasoning because the meaning of a record changes depending on timing, quantity, movement, and context. A high quantity may be manageable if the expiration window is long, while a smaller quantity requires immediate action if the expiration window is short or if the record violates expected first-expired-first-out behavior. The system needed to support evidence-based interpretation, not just display raw values.

Cloud management concepts (IST 615) helped me frame the application as a lightweight platform rather than a static report. The value came from giving users an accessible, low-friction review surface that could sit between source data and operational action. The system did not need to be a large enterprise deployment to matter. It needed to be available, usable, structured, and capable of turning operational data into decision-ready information.

The core design pattern was expiration-aware prioritization. The system assigned inventory into generalized priority bands so users could focus on what required the earliest review. That mattered because expiration-sensitive inventory does not create equal urgency across all records. The system had to separate items that were critical from items that simply needed monitoring. This helped convert a large inventory population into a more manageable action queue.

Exception detection was the second major design pattern. The system identified records that required review because of near expiration, unexpected movement, quantity concerns, location concerns, missing date information, or potential first-expired-first-out violations. Exception surfacing turned raw inventory data into a review workflow. The system brought likely issues forward automatically, while still needing to avoid the opposite problem of too many low-value signals. Decision-support research warns that alerts and recommendations must be integrated carefully into workflow or they can create fatigue rather than support.[14]

Filtering and segmentation made the system usable. Users could narrow the review by generalized categories such as location group, material category, priority band, review status, expiration window, or movement type. That self-service design is important because different stakeholders ask different questions. A leader may want to know the size of the risk. A planner may want to know which category is affected. A material handler may need to know which specific item requires action. The system gave those users a way to answer their own questions without repeatedly asking for a new extract.

The snapshot and replay logic also mattered. Inventory is inherently a time-based problem. A point-in-time snapshot preserves what the system knew at a particular moment, unlocking true risk and root cause analysis. By providing the ability to essentially "time travel" back to another day, the system recreates the exact environmental factors, showing exactly what

the inventory looked like at that time. This means users no longer have to guess; they can know for a fact *why* something happened. That supports review, learning, and operational continuity because it allows users to understand exactly how risk changed over time. This is part of what moved the project from a dashboard toward an intelligence surface.

Stakeholder design was again central. The system had to work for people operating under time pressure. It needed to be clear at a glance, detailed when needed, and structured enough to support follow-up. The design had to make the next action immediately clear, avoiding any need for users to understand the underlying transformation steps.

This project also shows how I use AI in a controlled way. The application was submitted for internal innovation recognition in the AI category, but the value was not blind automation. The value was using AI-assisted design and development to create a better operational intelligence surface while keeping the underlying logic structured, reviewable, and grounded in source data. AI helped accelerate the design and communication of the system, while the operational value came from the way the system organized inventory risk for human decision-making.

The strongest lesson from this project is that applied data science can improve operational judgment. The system structured the data so that action became clearer. In that sense, the Inventory Control System extended the same pattern that appears across my portfolio: build a shared truth layer, organize it around the user's decision, and create a system that reduces repeated questions by making the answer visible at the moment it is needed.

Cross-Course Synthesis

The three case studies operate in different domains, but they are not separate stories. The Financial Control System addressed reconciliation and financial risk. The Chemical Compliance System addressed regulatory review, accountability, and audit defense. The Inventory Control System addressed expiration-aware prioritization and operational intelligence. Each system solved a different problem, but the pattern underneath them was consistent:

1. Context
2. Structure
3. Modularity
4. Evidence
5. Iteration

That pattern is the clearest evidence of my development across the program. My coursework provided a framework to approach ambiguous operational problems and turn them into systems that other people could use, review, and trust.

The first step is context. In each project, the visible request rarely captured the full scope of the problem. In the Financial Control System, the visible request was reconciliation, but the deeper issue was whether the organization could explain variance in a repeatable and defensible way. In the Chemical Compliance System, the visible request was annual reporting, but the deeper issue was whether regulated information could be reviewed, trusted, and reconstructed under deadline pressure. In the Inventory Control System, the visible request was inventory visibility, but the deeper issue was whether users could understand which material needed attention first. Applied data science begins by recognizing that the first question is often just the surface layer.

The second step is structure. The foundation of systems analysis (IST 654) taught me to define users, requirements, boundaries, inputs, outputs, and process assumptions, while database management (IST 659) reinforced that reliable outputs depend on reliable structures. Across all three systems, structure became the difference between a useful file and a repeatable process. I had to define what belonged in scope, what relationships mattered, what evidence needed to be preserved, and how the workflow should behave when records were missing, mismatched, flagged, or incomplete.

The third step is modularity. A fragile process often collapses too many things into one place: source data, transformation logic, review decisions, stakeholder communication, and final output. My coursework helped me separate those concerns. In the Financial Control System, that meant separating financial billing behavior from physical inventory behavior, then recombining them into a stronger hybrid review model. In the Chemical Compliance System, it meant separating data gathering, review status, sign-off, and audit history. In the Inventory Control System, it meant separating raw inventory visibility from priority bands, exceptions, filters, and review actions. Modularity made the systems easier to explain, easier to improve, and easier to trust.

The fourth step is evidence. This may be the most important thread across the portfolio. My military background taught me that information has to be useful under pressure, and my coursework gave me better methods for preserving that usefulness in data systems. Evidence requires explaining where a final number came from, what assumptions shaped it, who reviewed it, when it changed, and whether it can be reconstructed later. That mattered in financial reconciliation, chemical compliance, and inventory risk because each domain required defensibility. The systems were designed to preserve the logic behind the answers. This is consistent with the broader data governance concern that trustworthy decision outcomes depend

on users understanding the quality, origin, and reliability of the information supporting the decision.[15]

The fifth step is iteration. None of these systems emerged fully formed. They developed through use, feedback, failure points, timing pressure, stakeholder questions, and repeated refinement. That is where the coursework and professional work reinforced each other most clearly. My workplace gave me real constraints: deadlines, fragmented systems, operational urgency, and people who needed the work to function. My coursework gave me language and methods for improving the work: requirements, governance, schema thinking, platform design, quantitative reasoning, and decision support. The result was an applied learning cycle rather than a one-way transfer from classroom to workplace.

The courses combined into a unified method. For example, a compliance workflow is simultaneously a governance problem (IST 614), a requirements problem (IST 654), an adoption problem (IST 621), and a traceability problem (IST 659). An inventory application spans business analytics, systems analysis, user workflow, time-based reasoning, and platform delivery. The real value of the program was learning to see these layers operating at the same time.

The cross-course lesson is that applied data science is fundamentally a translation discipline. It translates messy operational reality into structured information. It translates stakeholder questions into system requirements. It translates data into reviewable evidence. It translates one-time work into repeatable workflows. Most importantly, it translates learning into capability.

That is the central argument of this portfolio. My coursework extended beyond the classroom, and my work evolved past a collection of isolated tasks. Together, they formed a feedback loop that changed how I build. I became less focused on producing individual outputs and more focused on designing systems that remove entire categories of questions from the workflow by making trusted answers visible, self-service, and available when people need them.

Reflection and Future Direction

Beyond building three operationally deployed systems, the most important outcome of this portfolio is that I can now describe the professional method behind them. Before this program, I often understood my work through the immediate output: a report, a dashboard, a reconciliation file, a data extract, or a presentation. Those outputs still matter, but they are no longer how I define the work. I now understand my work as system-building.

That shift matters because systems carry responsibility. A report can answer a question once. A system can change how a team works, how risk becomes visible, how people trust

information, and how future questions are handled. Across the Financial Control System, the Chemical Compliance System, and the Inventory Control System, I learned that the value of applied data science relies on structure, traceability, usability, governance, and trust just as much as technical accuracy.

My military background shaped this development. In the Air Force, I learned that information has to work under pressure. It has to be clear, timely, and useful to the people responsible for acting on it. That lesson carried into my civilian work, but my graduate coursework gave me the formal language for it. My education provided the frameworks to define user boundaries, enforce data structure, navigate organizational culture, align with enterprise governance, and connect quantitative analysis directly to decisions under uncertainty.

This portfolio also changed how I understand service. Leaving the military did not end that part of my identity. It changed the form of it. In my current work, service means building systems that help other people make better decisions, reduce avoidable confusion, and spend less time reconstructing information that should already be available. The systems in this portfolio are not abstract exercises. They came from real operational pressure, real stakeholder needs, and real consequences.

The Financial Control System taught me that analysis becomes more powerful when it becomes control. The system had to make variance review more repeatable, more explainable, and more actionable. The Chemical Compliance System taught me that compliance work must outlast the person doing it. A process that depends on memory, scattered files, or last-minute reconstruction is inherently unstable for a regulated environment. The Inventory Control System taught me that visibility is only the first step. The stronger goal is operational intelligence: helping users understand what needs attention first and why.

A major part of my growth was learning to distinguish automation from system design. Automation can make a weak process faster, but it does not automatically make the process better. This caution is supported by decision-support research, which shows how flawed or poorly integrated support systems can harm trust and performance rather than improve them.[16] A stronger system clarifies the process before accelerating it. It defines what belongs in scope, what evidence matters, what review states mean, who needs to act, and how future users will understand the output. That is the standard I want to carry forward.

This also shaped my view of AI. I use AI as a thinking and design partner, keeping judgment, validation, and accountable logic strictly in human hands. In my work, AI can help draft language, compare approaches, organize ideas, accelerate development, and clarify communication. However, the systems themselves still need deterministic logic, structured data, documented assumptions, and human review. The more powerful AI becomes, the more important those disciplines become.

Professionally, I want to continue building production-grade, local data systems that sit between operational reality and strategic decision-making. I want to keep improving the way organizations use data to reduce ambiguity, preserve evidence, and support better decisions. I also want to continue developing systems that are self-service by design, where stakeholders can answer recurring questions without needing to rebuild the logic each time.

The central thesis remains the best way to close the paper:

"My work informs my degree, and my degree informs my work."

That feedback loop is no longer just a statement about school and work. It is the foundation of my professional identity. I am leaving this program with stronger technical skills, but more importantly, I am leaving with a clearer understanding of what I build and why I build it. I build systems that make information more trustworthy, work more understandable, and decisions more grounded in evidence.

Notes

1 Paul Brous and Marijn Janssen, "Trusted Decision-Making: Data Governance for Creating Trust in Data Science Decision Outcomes," Administrative Sciences 10, no. 4 (2020): 81, https://doi.org/10.3390/admsci10040081.

2 René Abraham, Johannes Schneider, and Jan vom Brocke, "Data Governance: A Conceptual Framework, Structured Review, and Research Agenda," International Journal of Information Management 49 (2019): 424-438, https://doi.org/10.1016/j.ijinfomgt.2019.07.008.

3 Philipp Brauner, Ralf Philipsen, André Calero Valdez, and Martina Ziefle, "What Happens When Decision Support Systems Fail? The Importance of Usability on Performance in Erroneous Systems," Behaviour & Information Technology 38, no. 12 (2019): 1225-1242, https://doi.org/10.1080/0144929X.2019.1581258.

4 Olufisayo Olusegun Olakotan and Maryati Mohd Yusof, "The Appropriateness of Clinical Decision Support Systems Alerts in Supporting Clinical Workflows: A Systematic Review," Health Informatics Journal 27, no. 2 (2021), https://doi.org/10.1177/14604582211007536.

5 Michael G. Alles, Gerard Brennan, Alexander Kogan, and Miklos A. Vasarhelyi, "Continuous Monitoring of Business Process Controls: A Pilot Implementation of a Continuous Auditing System at Siemens," International Journal of Accounting Information Systems 7, no. 2 (2006): 137-161, https://doi.org/10.1016/j.accinf.2005.10.004.

6 Marta E. Staff and Navonil Mustafee, "Discrete-Event Simulation for Effective Perishable Inventory Management: A Review," Simulation 101, no. 9 (2025): 981-1000, https://doi.org/10.1177/00375497251334387.

7 Brous and Janssen, "Trusted Decision-Making," 81.

8 Abraham, Schneider, and vom Brocke, "Data Governance," 424-438.

9 Alles et al., "Continuous Monitoring of Business Process Controls," 137-161.

10 Brauner et al., "What Happens When Decision Support Systems Fail?," 1225-1242.

11 Abraham, Schneider, and vom Brocke, "Data Governance," 424-438.

12 Olakotan and Yusof, "Appropriateness of Clinical Decision Support Systems Alerts."

13 Staff and Mustafee, "Discrete-Event Simulation," 981-1000.

14 Olakotan and Yusof, "Appropriateness of Clinical Decision Support Systems Alerts."

15 Brous and Janssen, "Trusted Decision-Making," 81.

16 Brauner et al., "What Happens When Decision Support Systems Fail?," 1225-1242.

Bibliography

Abraham, René, Johannes Schneider, and Jan vom Brocke. "Data Governance: A Conceptual Framework, Structured Review, and Research Agenda." *International Journal of Information Management* 49 (2019): 424-438. https://doi.org/10.1016/j.ijinfomgt.2019.07.008.

Alles, Michael G., Gerard Brennan, Alexander Kogan, and Miklos A. Vasarhelyi. "Continuous Monitoring of Business Process Controls: A Pilot Implementation of a Continuous Auditing System at Siemens." *International Journal of Accounting Information Systems* 7, no. 2 (2006): 137-161. https://doi.org/10.1016/j.accinf.2005.10.004.

Brauner, Philipp, Ralf Philipsen, André Calero Valdez, and Martina Ziefle. "What Happens When Decision Support Systems Fail? The Importance of Usability on Performance in Erroneous Systems." *Behaviour & Information Technology* 38, no. 12 (2019): 1225-1242. https://doi.org/10.1080/0144929X.2019.1581258.

Brous, Paul, and Marijn Janssen. "Trusted Decision-Making: Data Governance for Creating Trust in Data Science Decision Outcomes." *Administrative Sciences* 10, no. 4 (2020): 81. https://doi.org/10.3390/admsci10040081.

Olakotan, Olufisayo Olusegun, and Maryati Mohd Yusof. "The Appropriateness of Clinical Decision Support Systems Alerts in Supporting Clinical Workflows: A Systematic Review." *Health Informatics Journal* 27, no. 2 (2021). https://doi.org/10.1177/14604582211007536.

Staff, Marta E., and Navonil Mustafee. "Discrete-Event Simulation for Effective Perishable Inventory Management: A Review." *Simulation* 101, no. 9 (2025): 981-1000. https://doi.org/10.1177/00375497251334387.